Functional Description

MTU 8V4000 GS - 8V4000L64
MTU 12V4000 GS - 12V4000L64
MTU 16V4000 GS - 16V4000L64
MTU 20V4000 GS - 20V4000L64

MS13056/00E

Sales Name	Engine model	Application group
MTU 8V4000 GS	8V4000L64	3A, continuous operation, unrestricted
MTU 12V4000 GS	12V4000L64	3A, continuous operation, unrestricted
MTU 16V4000 GS	16V4000L64	3A, continuous operation, unrestricted
MTU 20V4000 GS	20V4000L64	3A, continuous operation, unrestricted

Table 1: Applicability

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1 Engine-Generator Set

1.1 General description

The genset comprises the following main assemblies:

- Engine
- Gas train
- Generator and coupling
- Baseframe
- Heat supply from cogeneration (with versions GR and GC)
- MMC (MTU Module Control) for plant control, closed-loop control, diagnostics (optional)
- MIP (MTU Interface Panel) for engine control, closed-loop control, diagnostics

In the combustion engine, the energy contained in the fuel is converted into mechanical work and heat.

The engine coolant dissipates the engine heat and the withdrawn heat quantity via a plate-core heat exchanger.

The generator, which converts the mechanical work into electrical energy, is flanged-mounted on the engine via a resilient coupling. The generator is installed on two slide rails, which allow the generator to be shifted when the coupling element is being replaced. The engine and generator are connected to the baseframe via resilient, vibration-damped elements.

All closed-loop controls, control systems and monitoring systems, including communication options to and from the system, are implemented in the hardware and software in the switch cabinet.

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1.2 Engine-generator set – Application group

The engine-generator set consists of a gas engine and a generator mounted on a common baseframe. Upon demand, the engine will start and drives the generator to produce electrical power.

Application group 3A – Continuous operation, unrestricted (Continuous Power)

Engine-generator sets for unlimited continuous power supply are used for:
- Base load generation
- Part of a power supply or the public grid

Continuous operation	Application group 3A
Operating mode	Continuous operation, unrestricted
Rating definition	10% overload capacity (ICXN)
Load factor	< 100 %
Operating hours	Unrestricted

Benefits

- Wide range of standardized engine-generator sets to meet customer requirements with regard to power, emission and other characteristics.
- Possibility to choose from different component designs (e.g. generator) and to select options (e.g. heat module)
- Most recent gas engine technology.
- Main components with state-of-the-art technology for high electrical and thermal efficiency and long service life.

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1.3 Engine-generator set – Nameplate and designation

Nameplates on engine-generator set

The genset has the following nameplates:
- Nameplate of engine-generator set
- Engine nameplate
- Nameplate of generator

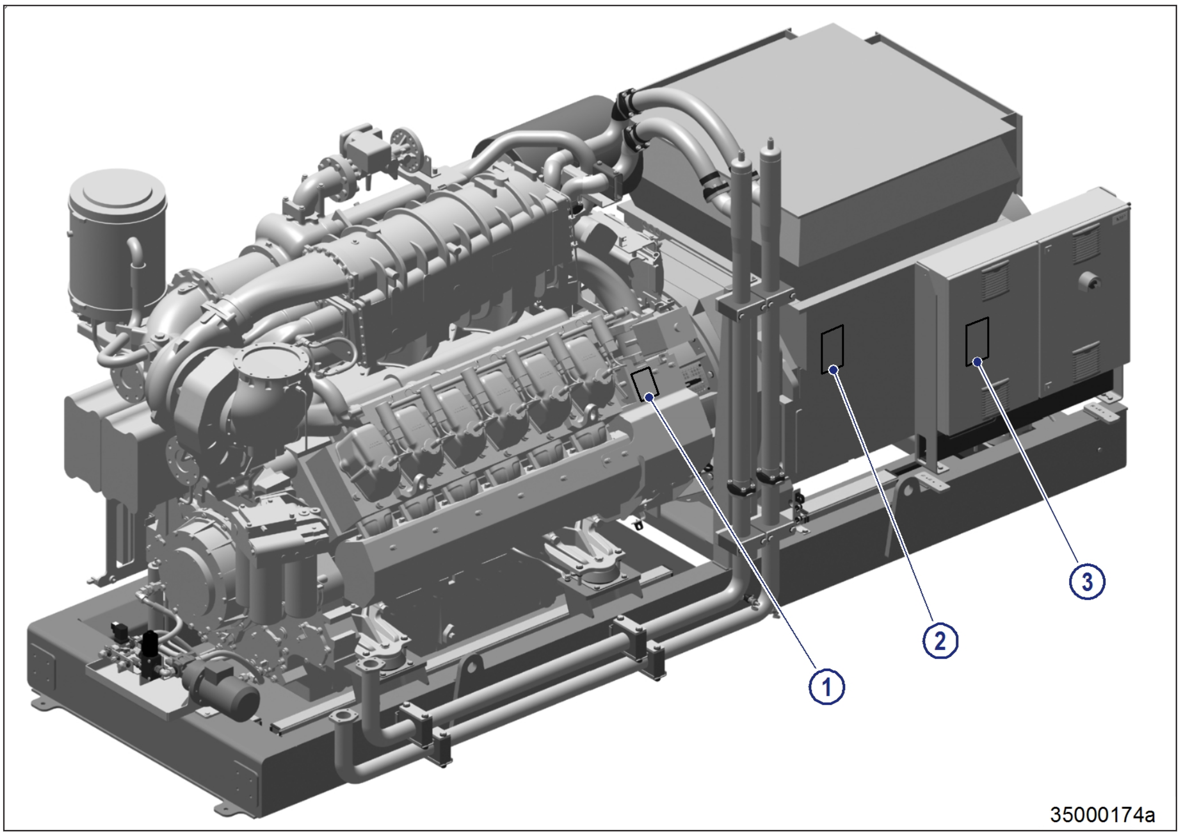

Figure 1: Fitting position of nameplates (generator and engine-generator set)

1 Engine nameplate
2 Nameplate of generator
3 Nameplate of engine-generator set

Explanation of product name

Example:MTU 12V4000 GS	
MTU	Manufacturer: MTU Onsite Energy
12	Number of cylinders of the engine
V	Cylinder arrangement of the engine Vee arrangement
4000	MTU Series 4000
GS	Gas System

Overview of application groups on nameplate

Specification on the nameplate	Application group
CONTINUOUS POWER	3A – Continuous operation, unrestricted
PRIME POWER	3B – Continuous operation, variable load, ICXN
LTP GRID STABILITY	3G – Continuous operation, time limited, ICXN

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Specification on the nameplate	Application group
DATA CENTER CONTINUOUS	3F – Standby operation mode, unlimited, ICXN
MISSION CRITICAL STANDBY	3H – Standby operation mode, special qualifications

Engine-generator set – nameplate and designation

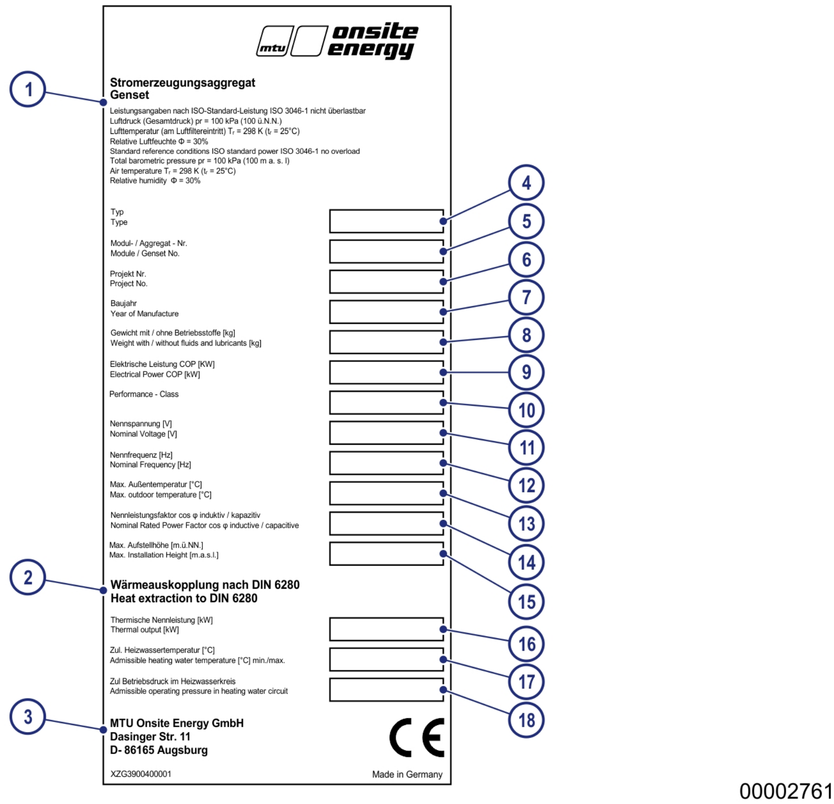

Figure 2: Nameplate of engine-generator set

Item	Designation
1	Power ratings as per ISO standard power ISO 3046-1 no overload capability Air pressure (total pressure) pr = 100 kPa (100 above sea level) Air temperature (at air filter inlet) T_r = 298 K (t_r = 25 °C) Relative air humidity ϕ = 30%
2	Heat supply from cogeneration as per DIN 6280
3	MTU Onsite Energy Dasinger Str. 11 D-86165 Augsburg
4	Type
5	Module / genset no.
6	Project no.
7	Year of construction
8	Weight with / without fluids and lubricants [kg]
9	Electrical output COP [KW]
10	Performance class
11	Rated voltage [V]
12	Rated frequency [Hz]
13	Max. outside temperature [°C]

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Item	Designation
14	Rated power factor cos ϕ inductive / capacitive
15	Max. site altitude [m above sea level]
16	Thermal rated power [kW]
17	Perm. heating water temperature [°C]
18	Perm. operating pressure in heating water circuit

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1.4 Engine-generator set – Standard scope of delivery

The illustrations show an engine-generator set featuring a Series 12V 4000 L64 engine. The illustrations also apply to engine-generator sets with 8, 16 and 20-cylinder engines.

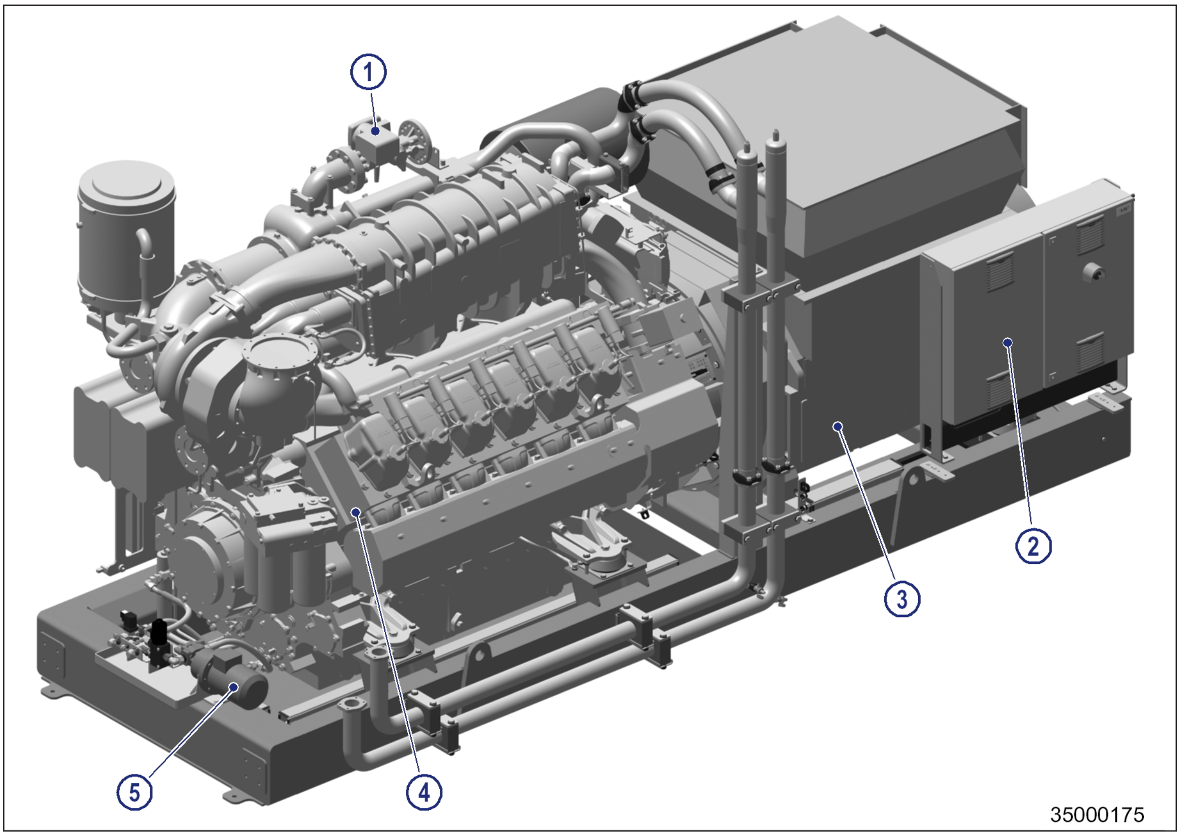

Figure 3: Engine-generator set – Engine end

1 Gas train	3 Generator	5 Lube oil pump
2 Switch cabinet	4 Engine	

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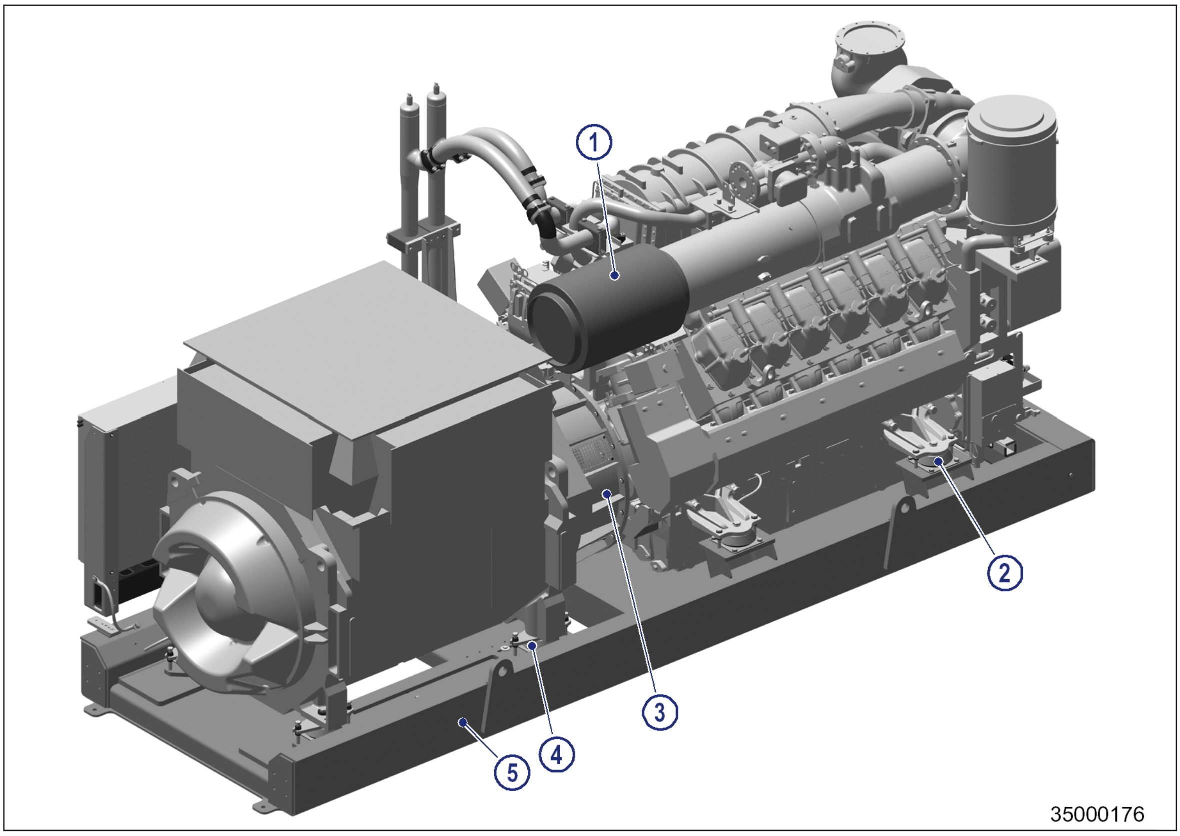

Figure 4: Engine-generator set – Generator end

1 Air filter	3 Coupling	5 Baseframe
2 Engine mounting	4 Generator mount	

Standard scope of delivery Genset Basic (GB)

- Gas engine of the MTU 4000 Series:
  - Rated speed 1,500 rpm (50 Hz)
  - Electronic engine management system (ECU)
  - Electric starter
- Generator:
  - Voltage 400 V, 50 Hz
  - 2 bearings
  - Digital voltage regulator
  - Measurement current transformer x/ 1A
  - Protection current transformer x/ 1A 5P10
  - Winding temperature monitor 2 PT100 per winding
  - Bearing temperature monitor 1 PT100 per bearing
  - Anti-condensation heating
  - Degree of protection IP23
- Gas system: Gas train with connecting hoses to genset
- Oil system with oil pump for priming and draining the oil pan
- MIP control system: Switch cabinet with control function (interface between engine-generator set and customer controller)
- Air filter with air pipework to gas mixer. Additional air filter housing for 20V engines
- Coupling
- Exhaust pipe bellows with further accessories (e.g. hoses, pole caps, damping plates, etc.)
- Safety guards for all rotating parts
- Baseframe with resilient engine and generator mounts

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